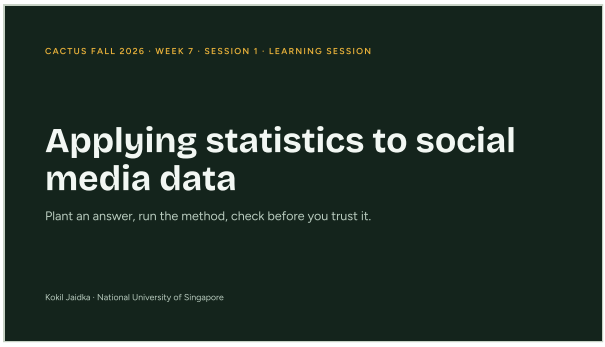
# CACTUS Week 7: Applying Statistics to Social Media Data




## Slide 1

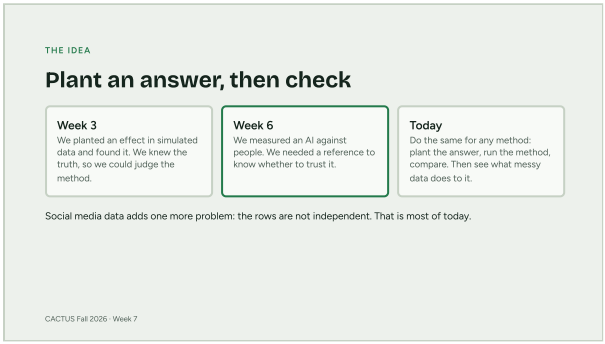
**TIME** 1 min (0:00)

**SAY** Welcome. Over the next two sessions we apply statistics to social media data. Today is the idea that everyone needs. In Session 2 you split into tracks.

**DO** Welcome people. Post the course site link. Say the plan: this learning session, the practice page, then a 2-hour coding session; Session 2 is the tracks.

**NOTE** This is about a 55-minute session. The hands-on work is in the coding session.

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## Slide 2

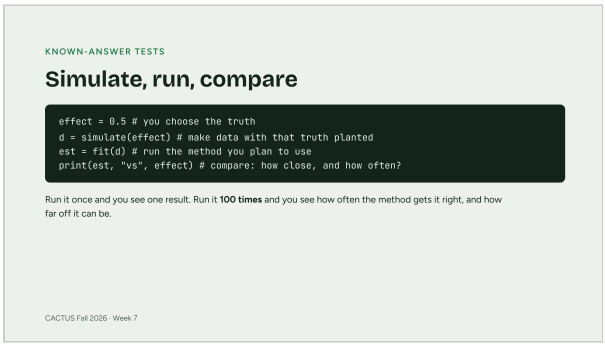
**TIME** 4 min (0:01 to 0:05)

**SAY** In Week 3 we planted an effect and found it. In Week 6 we measured an AI against people. Today we do the same for any method: plant the answer, run the method, compare. And then we deal with the problem that is specific to social media: the rows are not independent.

**DO** Read the three cards. Ask for a show of hands: who has analysed comments or posts before?

**IF ASKED whether we are doing multilevel modelling** say that we start with one practical fix (clustered standard errors) and show the mixed model as one line.

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## Slide 3

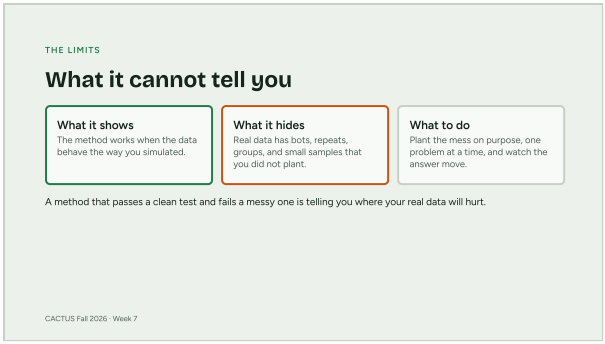
**TIME** 5 min (0:05 to 0:10)

**SAY** A known-answer test has three steps. Simulate data where you choose the truth. Run the method you plan to use. Compare. The code on the slide is the whole idea.

**DO** Ask the room to predict: if I run one simulated study, will the estimate equal the truth exactly? Take two answers. Then say what happens over 100 runs: it scatters around the truth. If you are comfortable, run a small version live in Colab; if not, describe what you would expect to see.

**WATCH FOR** People who think a good result on one run proves the method. Say that one run is one draw.

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## Slide 4

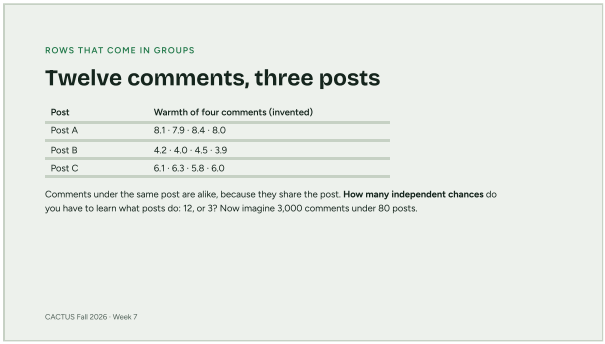
**TIME** 5 min (0:10 to 0:15)

**SAY** A known-answer test shows that a method works when the data behave the way you simulated. It cannot show that it works on real data. Real data has bots, repeats, groups, and small samples that you did not plant.

**DO** Read the three cards. Say: the good news is that you can plant the mess on purpose, one problem at a time, and watch the answer move. We will do that in the coding session and in Session 2.

**WATCH FOR** Students who hear this as 'simulation is useless'. Say that it is how we learn what to look for.

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## Slide 5

**TIME** 8 min (0:15 to 0:23)

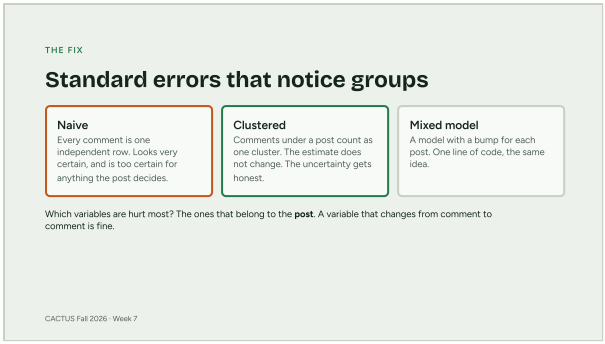
**SAY** Here are twelve comments under three posts. Look at the numbers. Comments under the same post are alike, because they share the post. So how many independent chances do you have to learn what posts do: twelve, or three?

**DO** Ask the room to predict first and take two answers. Three. Then say: now imagine 3,000 comments under 80 posts. If you analyse them as 3,000 independent rows, your result looks much more certain than it should.

**IF ASKED how to know whether rows are grouped** ask where the data came from. Comments in posts, posts by one account, people in one classroom or one condition: all of these are groups.

**WATCH FOR** Do not say what the notebook will find (the actual size of the effect). Students find that.

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## Slide 6

**TIME** 7 min (0:23 to 0:30)

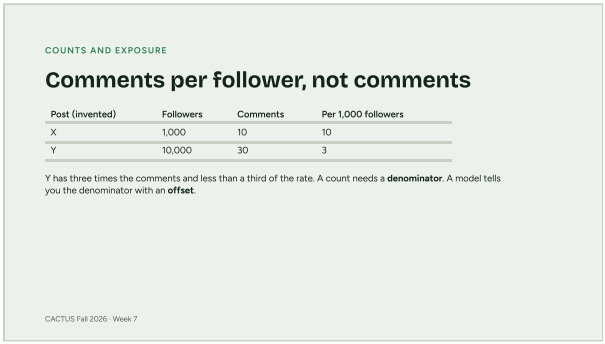
**SAY** There are three ways to handle it. The naive way treats every comment as independent. Clustered standard errors treat comments under a post as one cluster: the estimate does not change, but the uncertainty gets honest. A mixed model gives each post its own bump, in one line of code.

**DO** Say which variables are hurt most: the ones that belong to the post. A variable that changes from comment to comment, such as whether a comment is long, is fine. Say we will see this in the practice page and the notebook.

**IF ASKED which to use** say clustered errors first. The mixed model is a good next step and gives similar answers here.

**WATCH FOR** People who think clustering changes the estimate. It changes the standard error.

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## Slide 7

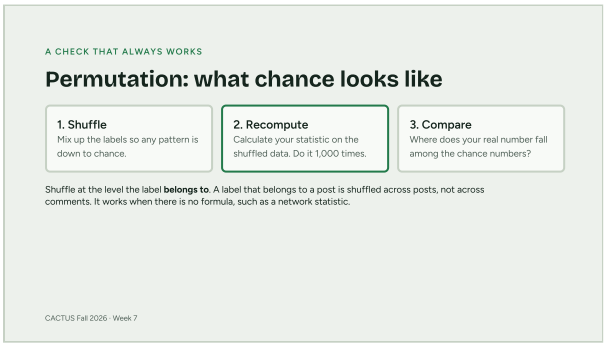
**TIME** 5 min (0:30 to 0:35)

**SAY** Post Y has three times the comments of X, but it has ten times the followers. Per 1,000 followers it has less than a third of the rate. A count needs a denominator. When you fit a model, you tell it the denominator with an offset.

**DO** Ask: which post is doing better, X or Y? Take two answers. Then ask what you would want to know before saying that.

**WATCH FOR** Students who think an offset is an extra predictor. It is a denominator that is built into the model, not estimated.

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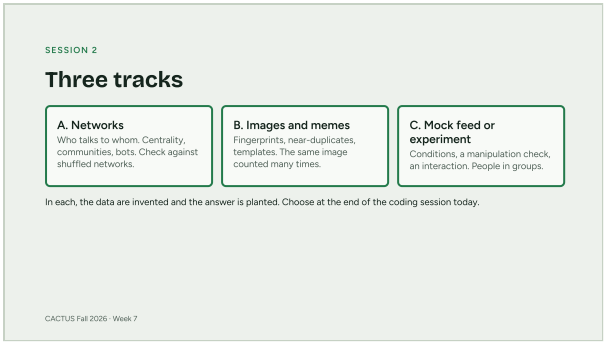

## Slide 8

**TIME** 5 min (0:35 to 0:40)

**SAY** A permutation test is a check that works when there is no formula. Shuffle the labels, so any pattern is down to chance. Recompute your statistic, 1,000 times. See where your real number falls among them.

**DO** Make the key point: shuffle at the level the label belongs to. If the label belongs to a post, shuffle it across posts, not across comments. Say that this is the exercise we will do at the end of the coding session.

**WATCH FOR** People who shuffle comments, because that is easier. Do not say what happens; they will see it.




## Slide 9

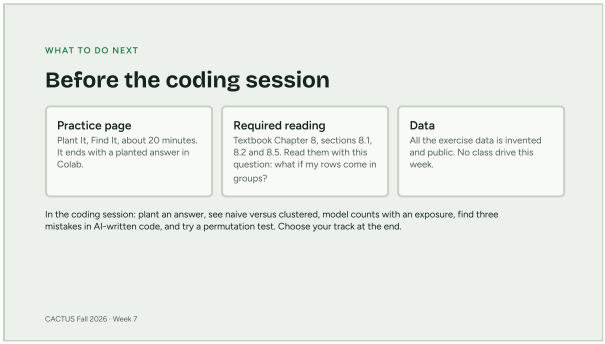
**TIME** 6 min (0:40 to 0:46)

**SAY** In Session 2 you pick one of three tracks. Networks: who talks to whom, with a hub and some bots planted. Images and memes: fingerprints and templates, with reposts planted. A mock feed or experiment: conditions, a manipulation check, and an interaction. Each has invented data with a planted answer, and each has its own way for the core ideas to show up.

**DO** Spend about two minutes on each. For networks, say the shuffled-network check is a permutation test. For images, say the same image counted many times is the repeat problem. For experiments, say people in groups is the grouping problem. Tell them to choose at the end of the coding session.

**WATCH FOR** Students unsure which track. Ask what their data look like: edges, images, or a comparison of conditions.

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## Slide 10

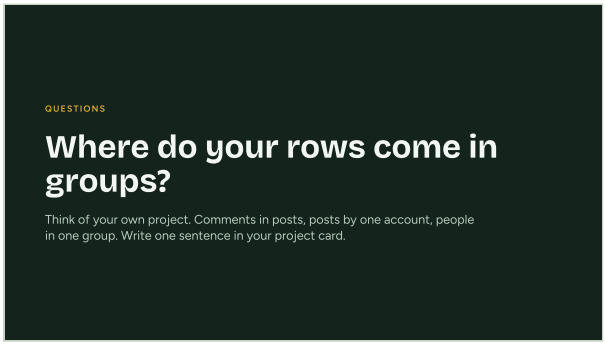
**TIME** 4 min (0:46 to 0:50)

**SAY** Before the coding session: the practice page, Plant It, Find It, about 20 minutes, and the required reading: textbook Chapter 8, sections 8.1, 8.2 and 8.5, read with the question 'what if my rows come in groups?'.

**DO** Post the practice page link and the course site link. Say that all the data this week is invented and public, so there is no class drive. Say that there is no lecture in the coding session, and that you will choose your track at the end.

**SAY** In the coding session: plant an answer, compare naive and clustered, model counts with an offset, find three mistakes in AI-written code, and try a permutation test.

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## Slide 11

**TIME** 4 min (0:50 to 0:54)

**SAY** Think of your own project. Where do your rows come in groups? Comments in posts, posts by one account, people in one group. Write one sentence in your project card.

**DO** Take questions. If you do not know the answer, say so and write it down.

**CLOSE** Thank everyone. Say the next part is the practice page, then the coding session.